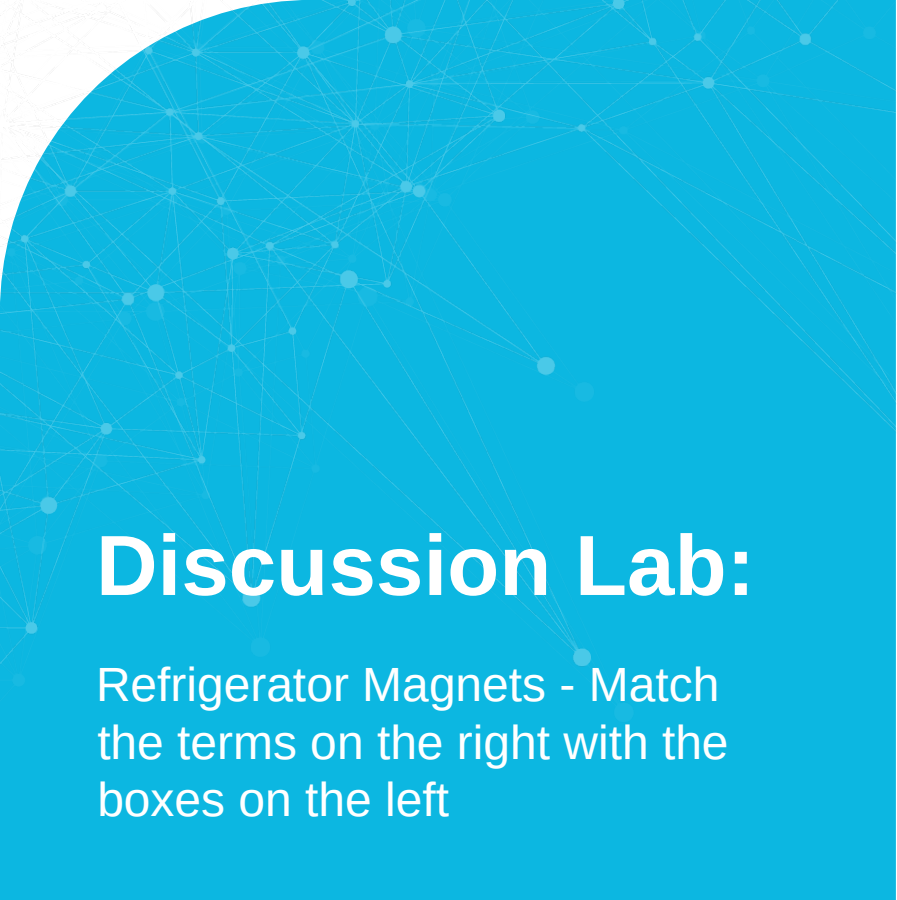


Discussion Unit:

DSE Ops Center

Introduce the DSE Operations Center

- Adding and expanding clusters
- Configuring nodes
- Viewing performance metrics
- Correcting issues
- Monitoring cluster health
- Configure security
- Backup and recovery
- Other

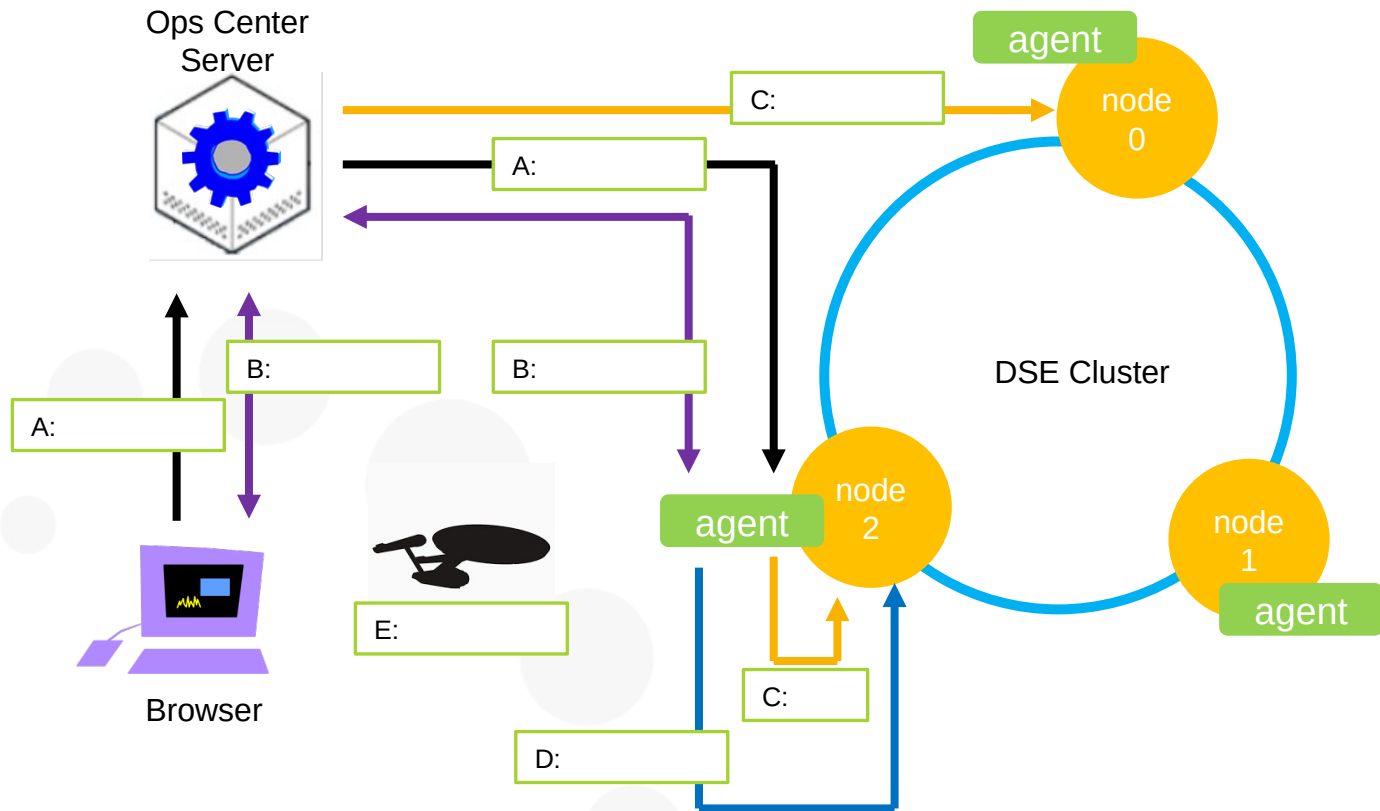


DSE, DSE Ops Center
Communication
Protocols/Architecture

Discussion Lab:

Refrigerator Magnets - Match
the terms on the right with the
boxes on the left

DSE, Ops Center: Communication Protocols/Arch

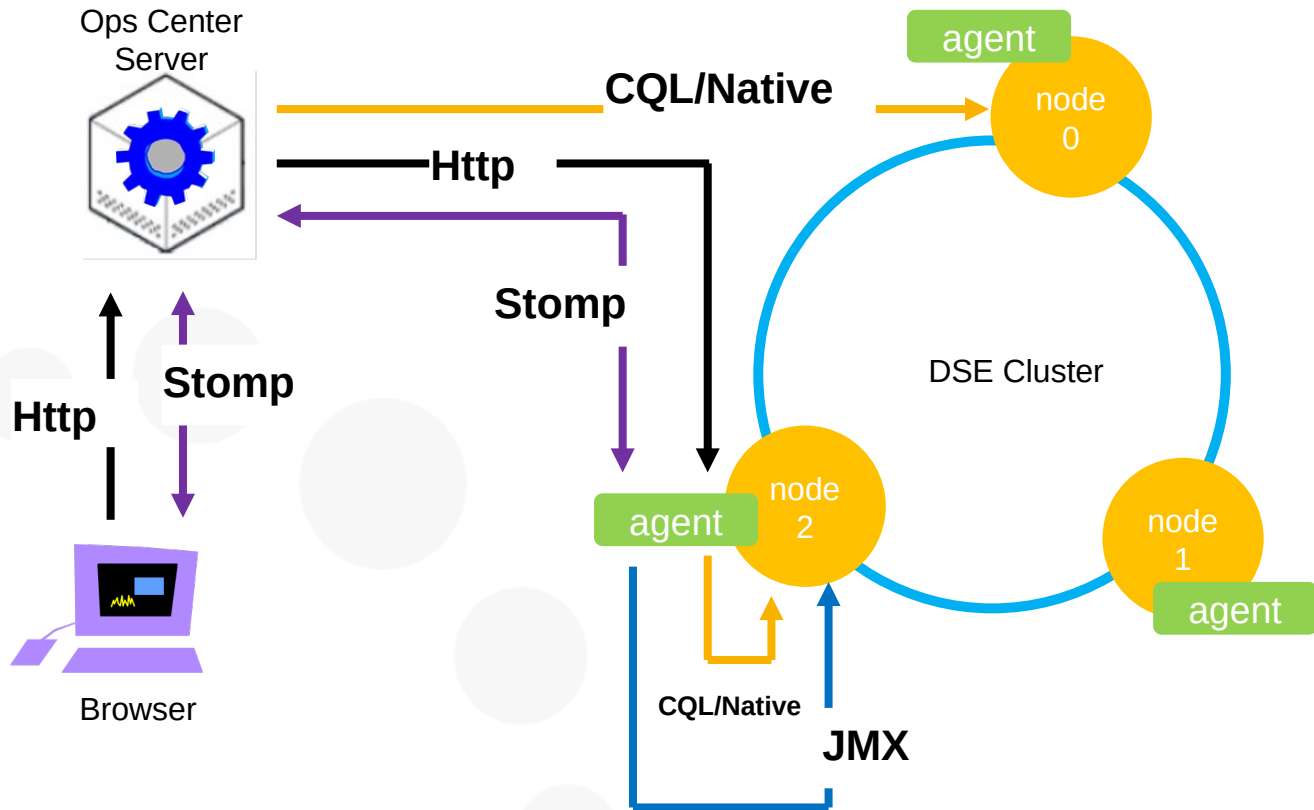


- JMX
- ODBC
- Http
- Stomp
- Sec-28, Sub-D
- CQL/Native



End of Discussion Lab:

Ops Center Communication Architecture



Ops Center Install Methods

Ops Center installation methods include the following:

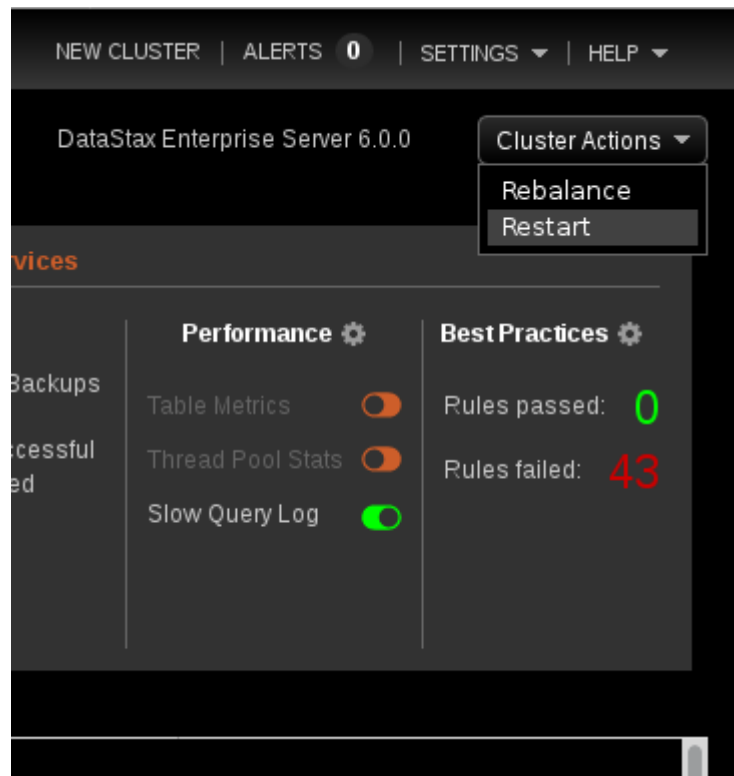
- Tar ball install (Practice Lab 6251 to follow)
- Docker install
- Linux Yum/APT-GET



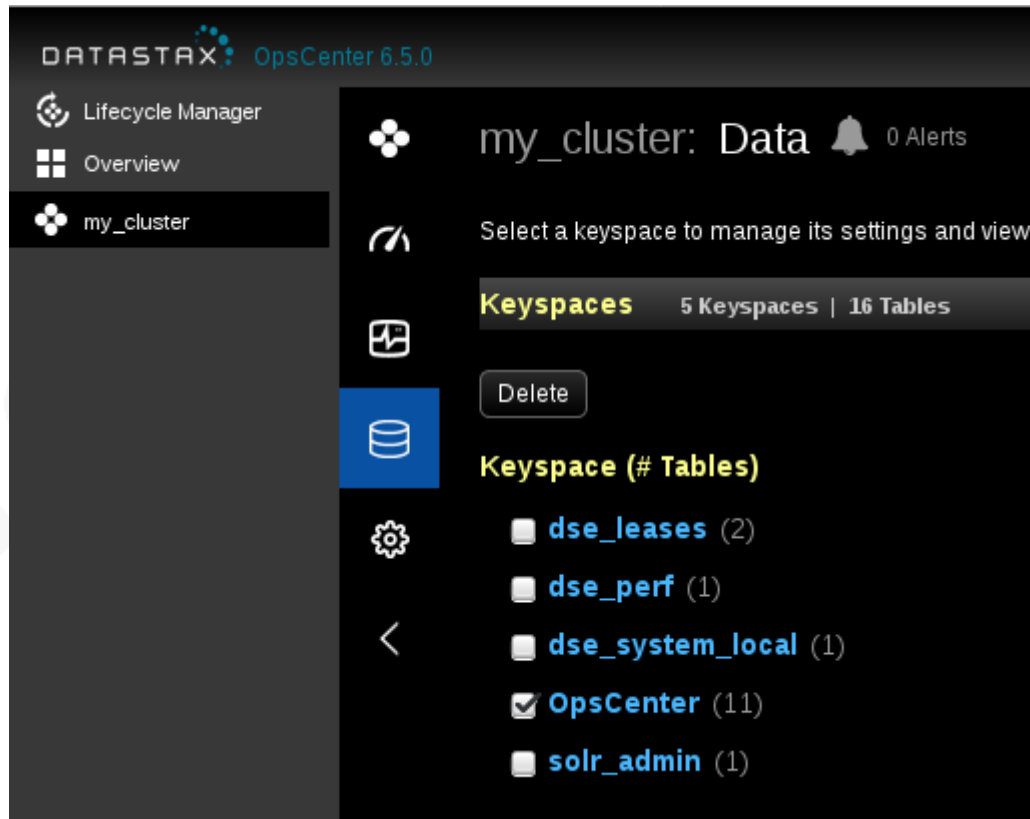
Ops Center: Cluster Ops

Ops Center capabilities relative to clusters include:

- Add an existing cluster, or create a new cluster
- Disconnecting an existing cluster from Ops Center and Lifecycle Manager
- Rebalancing a cluster
- Restarting a cluster
- Generating diagnostic data for a cluster
- Other



Ops Center: Keyspace Ops



Ops Center capabilities relative to keyspaces include:

- View/change keyspace settings
- Delete keyspaces
- Other

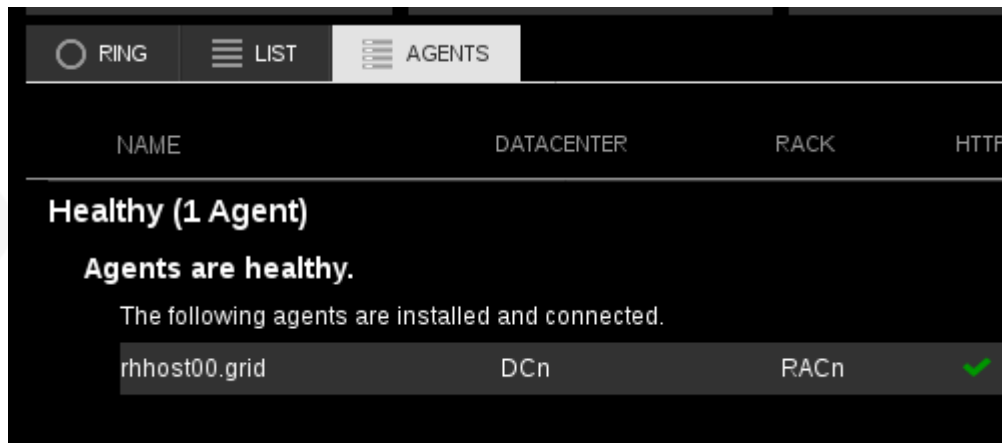
Ops Center: Table Ops

Ops Center capabilities relative to tables include:

- View table metrics
- Delete a table
- Truncate a table
- Other

The screenshot displays the DSE Ops Center interface. At the top, there's a navigation bar with a 'Back to Keyspaces' link. Below it, the 'dse_perf' keyspace is selected. The 'Keyspace Settings' section shows 'replica_placement_strategy' as 'org.apache.cassandra.locator.SimpleStrategy' and 'replication_factor' as '1'. Below the settings, there's a tabbed interface with 'TABLES' selected. Under the 'TABLES' tab, there's a 'Delete' button and a checkbox for 'node_slow_log'. To the right of the table list, the 'node_slow_log' table is highlighted, showing 'View Metrics' and 'Truncate' buttons. Below these buttons, the table's schema is displayed: 'CREATE TABLE dse_perf.node_slow_log (node_ip inet, date timestamp, ...'.

Ops Center: Agent Ops



The screenshot shows the 'AGENTS' tab in the Ops Center interface. At the top, there are three tabs: 'RING', 'LIST', and 'AGENTS'. Below the tabs, there is a table with columns: 'NAME', 'DATACENTER', 'RACK', and 'HTTP'. The table shows one agent, 'rhh00.grid', in the 'DCn' datacenter and 'RACn' rack, with a green checkmark in the 'HTTP' column. Above the table, there is a status message: 'Healthy (1 Agent)' and 'Agents are healthy. The following agents are installed and connected.'

NAME	DATACENTER	RACK	HTTP
Healthy (1 Agent)			
Agents are healthy.			
The following agents are installed and connected.			
rhh00.grid	DCn	RACn	✓

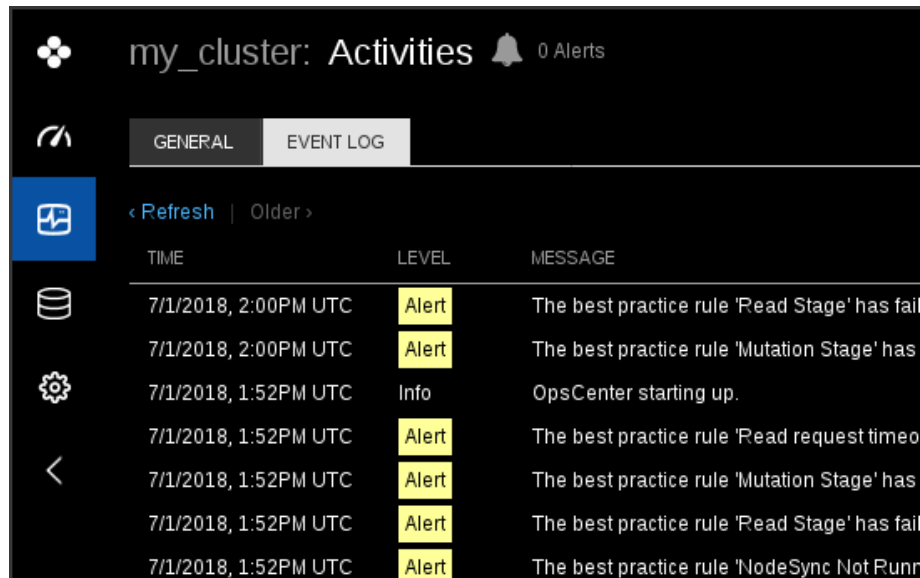
Ops Center capabilities relative to agents include:

- View agent status
- Install/upgrade agents
- Other

Ops Center: Alerts/Events

Ops Center capabilities relative to alerts/events include:

- Set logging level per; 0/Debug thru Alert/5.
- Specify SNMP alert parameters
- Specify SMTP alert parameters
- Specify Http/POST alert parameters
- Other



TIME	LEVEL	MESSAGE
7/1/2018, 2:00PM UTC	Alert	The best practice rule 'Read Stage' has fai
7/1/2018, 2:00PM UTC	Alert	The best practice rule 'Mutation Stage' has
7/1/2018, 1:52PM UTC	Info	OpsCenter starting up.
7/1/2018, 1:52PM UTC	Alert	The best practice rule 'Read request timeo
7/1/2018, 1:52PM UTC	Alert	The best practice rule 'Mutation Stage' has
7/1/2018, 1:52PM UTC	Alert	The best practice rule 'Read Stage' has fai
7/1/2018, 1:52PM UTC	Alert	The best practice rule 'NodeSync Not Runn

Ops Center capabilities relative to performance metrics include:

- Cluster wide performance metrics (metrics aggregated atop nodes)
- Pending task metrics; compactions, other
- Table performance metrics
- Tiered storage performance metrics
- Message latency metrics
- Search performance metrics
- Graph metrics
- Nodesync metrics
- Thread pool metrics
- Dropped messages metrics
- OS performance metrics
- Alert metrics

Ops Center: Performance Metrics

Ops Center: Management Services

	SERVICE	DESCRIPTION	STATUS	
	Backup Service	Schedule and restore backups (local or cloud). Also supports commit log backup for point in time restore.	SCHEDULED: 0	
	NodeSync Service	NodeSync is a data validation and syncing service managed by DataStax Enterprise. NodeSync must be configured in DSE. NodeSync status can be viewed in OpsCenter after enabling keyspaces and tables.	ON	
	Repair Service	Manage automatic repairs of all clusters; keeps data consistent on all nodes without impacting performance.	OFF	
	Best Practice Service	Automatically scans clusters using a set of pre-defined best practice rules and provides expert advice on resolving issues.	PASS: 11	FAIL: 32
	Capacity Service	Automatically collects statistical data for trend analysis and forecasts future trends.	ON	
	Performance Service	Combines OpsCenter metrics with CQL-based diagnostic tables populated by the DSE Performance Service to help understand, tune, and optimize cluster performance. Visually enable the performance objects and analyze the results within OpsCenter. Performance Service is available in DataStax Enterprise version 4.6 and higher.	ON	

Ops Center: Best Practice Rules Reference

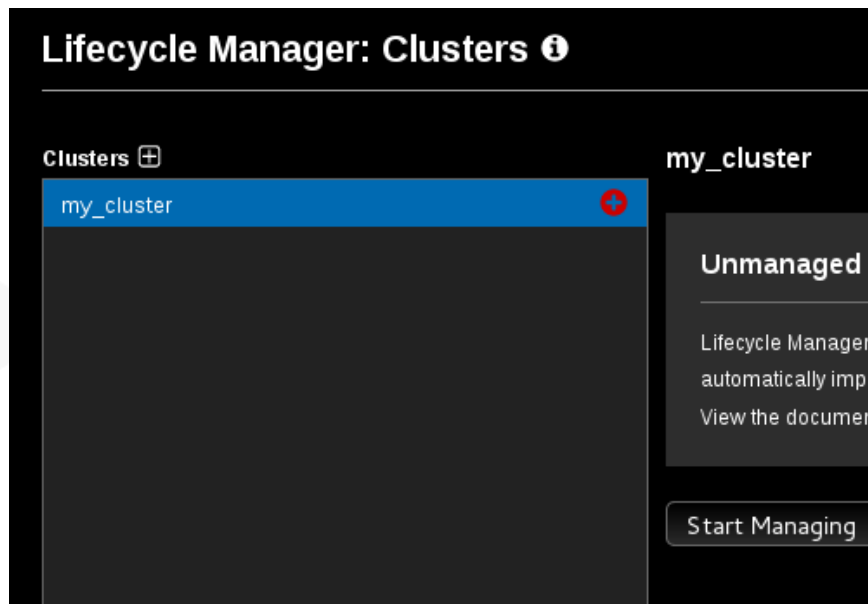
Easily 40+ Rules; Examples ..

- Checks that the default cassandra superuser and password has been changed from the default.
- Checks that you are not using SimpleStrategy for any keyspaces in a multi-datacenter environment.
- Checks to make sure SimpleSnitch isn't used in production.
- Checks for secondary indexes with too many distinct values.
- Others



Maury Atwater, President
of Atwater's

Ops Center: Lifecycle Manager (LCM)



Add Cluster

.....

Name *

A cluster name cannot be edited after creation.

SSH Credentials

SSH Management Port

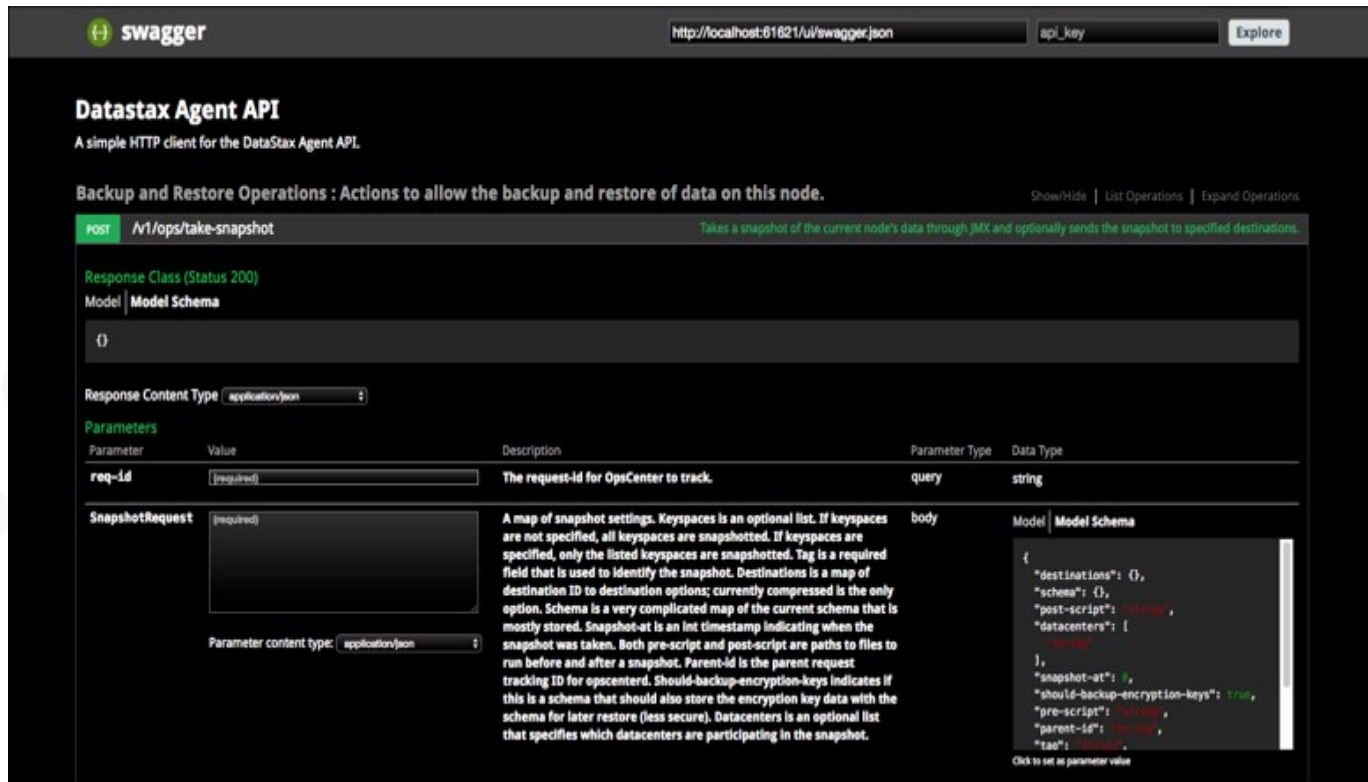
Repository

Config Profile

Select a config profile for the cluster so LCM can determine its authentication protocol and prompt for credentials upfront if necessary. Warning: If authentication is enabled and a config profile is not set at the cluster level, the credentials must be set manually using the LCM API.

Comment

Ops Center: Agent API



The image shows the Swagger UI for the Datastax Agent API. The URL is `http://localhost:61621/ui/swagger.json` and the API key is `api_key`. The API is titled "Datastax Agent API" and is described as "A simple HTTP client for the DataStax Agent API". The main section is "Backup and Restore Operations : Actions to allow the backup and restore of data on this node." with links for "Show/Hide", "List Operations", and "Expand Operations". The selected operation is `POST /v1/ops/take-snapshot` with the description "Takes a snapshot of the current node's data through JMX and optionally sends the snapshot to specified destinations." The response class is `Status 200` and the model is `Model Schema`. The response content type is `application/json`. The parameters table shows `req-id` as a query parameter and `SnapshotRequest` as a body parameter. The `SnapshotRequest` parameter is described as a map of snapshot settings, including destinations, schema, post-script, datacenters, snapshot-at, should-backup-encryption-keys, pre-script, parent-id, and tag. The parameter content type is `application/json`. A model schema is shown on the right, detailing the structure of the `SnapshotRequest` object.

swagger

`http://localhost:61621/ui/swagger.json` `api_key` [Explore](#)

Datastax Agent API

A simple HTTP client for the DataStax Agent API.

Backup and Restore Operations : Actions to allow the backup and restore of data on this node. [Show/Hide](#) | [List Operations](#) | [Expand Operations](#)

POST `/v1/ops/take-snapshot` Takes a snapshot of the current node's data through JMX and optionally sends the snapshot to specified destinations.

Response Class (Status 200)
Model | Model Schema

`{}`

Response Content Type `application/json`

Parameters

Parameter	Value	Description	Parameter Type	Data Type
<code>req-id</code>	<code>[required]</code>	The request-id for OpsCenter to track.	query	string

SnapshotRequest `[required]`

A map of snapshot settings. Keyspaces is an optional list. If keyspaces are not specified, all keyspaces are snapshotted. If keyspaces are specified, only the listed keyspaces are snapshotted. Tag is a required field that is used to identify the snapshot. Destinations is a map of destination ID to destination options; currently compressed is the only option. Schema is a very complicated map of the current schema that is mostly stored. Snapshot-at is an int timestamp indicating when the snapshot was taken. Both pre-script and post-script are paths to files to run before and after a snapshot. Parent-id is the parent request tracking ID for opscenterd. Should-backup-encryption-keys indicates if this is a schema that should also store the encryption key data with the schema for later restore (less secure). Datacenters is an optional list that specifies which datacenters are participating in the snapshot.

Parameter content type: `application/json`

Model | Model Schema

```
{
  "destinations": {},
  "schema": {},
  "post-script": "false",
  "datacenters": [
    "dc1"
  ],
  "snapshot-at": 0,
  "should-backup-encryption-keys": true,
  "pre-script": "false",
  "parent-id": "parent",
  "tag": "snapshot"
}
```

Click to set as parameter value

Edit the agent
address.yaml

swagger_enabled:
true

Restart agent

localhost:61621/ui

